Efficacy and effectiveness of therapist-guided internet versus face-to-face cognitive behavioural therapy for depression via counterfactual inference using naturalistic registers and machine learning in Finland: a retrospective cohort study

Tom H Rosenström, Suoma E Saarni, Samuli I Saarni, Jaakko Tammilehto, Jan-Henry Stenberg

Summary

Background According to meta-analyses of randomised controlled trials (RCTs), therapist-guided internet-delivered cognitive behavioural therapy (iCBT) is as effective a treatment for depression as traditional face-to-face CBT (fCBT), despite its substantially lower costs. However, RCTs are not always representative of routine practice, which could inflate effectiveness estimates. We combined rich data with counterfactual causal statistical reasoning to provide an fCBT–iCBT comparison complementary to RCTs.

Methods In this retrospective cohort study, we linked full archived therapist-guided iCBT and fCBT cohort registries with multiple Finnish social and health care registries. The therapist-guided iCBT programme with third-wave principles and the fCBTs were provided by HUS Helsinki University Hospital to people with depression without acute suicide or substance-misuse risk; fCBT was delivered in the Uusimaa region, whereas therapist-guided iCBT was nationwide and excluded people with treatment-interfering psychotic, neurological, or personality disorders, chronic or bipolar depression, or aged under 16 years. The primary outcome was the causal average treatment effect (ATE) for the difference in during-treatment symptom reductions between fCBT and therapist-guided iCBT, with symptoms measured by the Patient Health Questionnaire-9 (PHQ-9). If only one PHQ-9 report was recorded, no change was recorded, thereby penalising dropout. For an optimal and robust (a posteriori-balanced) ATE estimate, we applied targeted maximum likelihood machine learning. There was no involvement of individuals with lived experience in the research and writing process.

Findings The guided iCBT registry recorded patients from Dec 12, 2018, to Dec 22, 2022, and the fCBT registry spanned Aug 28, 2018, to Sept 28, 2022. From the total of 32 343 registered therapies, 392 people were included from the fCBT registry and 5467 people from the iCBT registry. Four people in the fCBT group and 21 in the guided iCBT group had missing baseline data, therefore the main sample for analysis contained 5834 patients (4101 [70%] were female and 1733 [30%] were male) with a mean age of 35 years (SD 12). Altogether, 5455 (94%) patients were registered with Finnish as their native language. The ATE estimate indicated that the PHQ-9 score declined 0·745 points (95% CI 0·156–1·334) more in the iCBT group than in the fCBT group. Sensitivity analyses concurred.

Interpretation Considered alongside previous RCTs, our findings suggest that short first-line treatments with therapist-guided iCBT are at least as effective and efficacious as fCBT. Our findings eliminate error sources and extend the representativeness of the population for this cost-effective treatment.

Funding The Research Council of Finland (Academy of Finland).

Copyright © 2025 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Lancet Psychiatry 2025; 12: 189–97

Published Online February 12, 2025 https://doi.org/10.1016/S2215-0366(24)00404-8

See Comment page 163

For the Finnish translation of the abstract see Online for appendix 1

Department of Psychology, Faculty of Medicine, University of Helsinki, Helsinki, Finland (T H Rosenström PhD, J Tammilehto PhD); **Department of Psychiatry, Helsinki University Hospital, Helsinki, Finland** (Prof S E Saarni PhD MD, J-H Stenberg PhD); **Faculty of Medicine and Health Technology, Tampere University, Tampere, Finland** (Prof S E Saarni, Prof S I Saarni PhD MD); **Family and Social Services, Psychiatry, Wellbeing Services County of Päijät-Häme, Lahti, Finland** (Prof S E Saarni); **Department of Psychiatry, The Wellbeing Services County of Pirkanmaa, Tampere, Finland** (Prof S I Saarni)

Correspondence to: Dr Tom H Rosenström, Department of Psychology, Faculty of Medicine, University of Helsinki, University of Helsinki, Helsinki 00014, Finland **tom.rosenstrom@helsinki.fi**

Introduction

Meta-analyses of available randomised controlled trials (RCTs) indicate that therapist-guided, internet-delivered cognitive behavioural therapy (iCBT) is a similarly effective treatment compared with the more costly and traditional face-to-face CBT (fCBT) for psychiatric disorders.[1,2] Much of the cost differential between therapist-guided iCBT and fCBT derives from therapists' differential time investment per patient. However, a meta-analysis[3] found no association between contact time and treatment effect in psychotherapy for depression. When people are directed to digital therapy supported by a therapist, their expectations of benefit are comparable to face-to-face therapy.[4] People with depression might benefit from therapist-guided iCBT and fCBT regardless of their symptom severity at intake.[5,6] However, some clinicians might find it difficult to accept the idea that

Research in context

Evidence before this study

Multiple meta-analyses indicate that cognitive behavioural therapy (CBT) is equally efficacious in treating depressive disorders when implemented as guided internet-delivered CBT (iCBT) compared with when delivered as the usual face-to-face CBT (fCBT). Because guided iCBT programmes use automated videos and exercises, online therapists can spend much less time per patient compared with face-to-face therapists, which can lead to significant costs savings. However, the meta-analyses rely on randomised controlled trials (RCTs), and the possibility remains that the trials do not represent the full routine-treatment population, or adequately account for the large differences in treatment adherence or other differences between fCBT and guided iCBT. Over the past few decades, the use of complementary sources of causal information has been increasingly recommended for all of medicine. One of the state-of-the-art methods for such causal triangulation is the targeted maximum likelihood estimation with machine learning. We searched PubMed (MEDLINE), with no language restrictions, for papers published from database inception to Sept 16, 2024, using the search terms: "targeted maximum likelihood" AND depression AND ("internet" OR "digital") AND ("cognitive behavioural therapy" OR "cognitive behavioral therapy"), yielding no results. A second search, replacing terms "targeted maximum likelihood" with "causal triangulation" or with "causality", or with "quasi-experiment", also gave no results. Overall, causal triangulation of the fCBT–iCBT efficacy differential in depression treatment is important given the substantial societal and monetary costs associated with depressive disorders and their treatment, which jointly imply a need to comprehensively understand the health effects of seemingly low-cost treatment policies.

Added value of this study

To our knowledge, this study is the first to evaluate quantitatively the efficacy differential between fCBT and guided iCBT in a format that is not an RCT or a meta-analysis of RCTs. We used counterfactual causal inference using targeted maximum likelihood machine learning and a unique linked registry from five distinct registry owners, combined using Finnish laws and regulations on secondary use of health and social registries. This targeting method can integrate features from both parametric and non-parametric methods to provide precise and reliable estimates of causal effects in situations in which the data are complex and could contain multiple confounding factors. We observed that the results from RCTs held true both in routinely collected data and in a posteriori-balanced (ie, causally informative) data: the guided iCBT programme was as effective and efficacious as fCBT in terms of depression symptom decline from before to after treatment (or to the last available measurement).

Implications of all the available evidence

Therapist-guided iCBT is equally effective and more cost-effective in treating depression than face-to-face delivery. Internet-delivery of short first-line treatments is a sensible default-treatment option and an apt response to the high demand for services.

partly automated treatments could yield treatment outcomes that are as effective as face-to-face therapies. The risk is that clinicians might inadvertently impart their attitudes to patients, potentially leading to diminished treatment efficacy.[4]

RCTs are not always representative of routine practice, which might inflate effectiveness estimates.[7,8] For example, people who object to digital treatments might not enrol in an RCT offering iCBT. Furthermore, samples in RCTs tend to be small relative to registries. Although meta-analyses combine RCTs to achieve larger samples, they often lack commensurate and comprehensive data on the complexity of people's conditions and other background factors that could affect representativeness. Thus, whether therapist-guided iCBT truly is as effective as fCBT in real-world settings remains unclear. The central counterfactual question is: would the average treatment effect (ATE) still be similar if we could allocate exact replicas of the same patients simultaneously to both guided iCBT and fCBT and compare their efficacies?

Although considered a gold standard, RCTs are neither the only nor a perfect method for balancing intervention group composition to avoid confounding. Researchers in medicine, psychology, and statistics argue that reliable causal inference often requires multiple sources of evidence rather than RCTs exclusively.[9–11] Many methods make assumptions that reduce bias from confounding (Figure 1). For example, estimates weighted by known inverse probabilities of treatment are free from confounding.[12,13] Alternatively, if the outcome can be accurately modelled as a function of confounders and treatment allocation, an unbiased estimate for treatment effect is achievable via G-computation.[14] If models for both outcome and treatment-assignment probability exist, models can be combined to achieve an augmented inverse-probability-of-treatment weighted (AIPTW) estimate that is doubly robust. This method ensures that a consistent estimate of the ATE is obtained if at least one of the models is correctly specified.[14] Finally, an ensemble of machine learners (ie, super learners) maximise the probability of at least one model being correct.[13,14]

In our case, the aforementioned methods make the working assumption that treatment outcome depends on the treatment (ie, delivery method). Outcome and treatment might depend on covariates, but all such confounding covariates are assumed to be fully observed. Other key assumptions are consistency

(ie, no other versions of the treatments), no interference (ie, a patient's outcome does not depend on the other patients), and positivity (ie, a positive probability for ending up in both iCBT and fCBT for all predictor covariate values). Finally, one should accurately capture forms of the dependencies indicated by the arrows in figure 1. Some methods need accuracy for both A and B dependencies, whereas doubly robust methods can achieve consistent estimates with only As or B and A_2 correctly modelled. Even methods not doubly robust might be preferred over naive regression analysis approaches, which are biased whenever the non-flexible parametric model is mis-specified relative to reality. Even application of mis-specified models that miss some covariates tends to provide more accurate estimates than naive approaches that estimate ATE with simple regression or mean difference.

A targeted maximum likelihood estimator (TMLE) based on super learners has the asymptotic double-robustness property, but it is additionally efficient, well defined, and well behaved in finite samples.[13,14] A TMLE integrates features from modern statistics, machine learning, and causal inference. A TMLE can optimise the bias-variance trade-off for a target parameter (eg, ATE) instead of the treatment-prediction and outcome-prediction models.[13,14] The primary motivation to use a TMLE is its ability to combine multiple desirable properties, perhaps representing the state of the art in covariate-based counterfactual causal inference. The availability of such procedures enhances the value of administrative, population-based registries, which often contain abundant aetiological information.[15]

Developments in Nordic countries enable the linking of psychotherapy data with information in national registries.[16–18] In this study, we used TMLE on these rich information sources to estimate the efficacy difference between therapist-guided iCBT and fCBT in a causally informative design that is not an RCT, thus circumventing typical challenges of RCTs while providing evidence on the same causal question. Our naturalistic data inform about effectiveness, and the counterfactual analysis informs about both effectiveness and efficacy if all real patients would receive fCBT versus guided iCBT. The new gold standard for evidence will then consist of combined inferences from RCT meta-analyses and TMLE applied to naturalistic datasets. Our investigation serves as a pragmatic proof of concept for the use of TMLE and related methods to address the question of the ATE between fCBT and therapist-guided iCBT.

Methods

Study design and participants

The study participants of this counterfactual inference-based cohort study were derived from naturalistic hospital treatment registries—the HUS Helsinki University Hospital iCBT (HUS-iCBT) registry and the Finnish Psychotherapy Quality Register (FPQR).[16] The iCBT that people received for depression in the HUS-iCBT registry largely corresponds to a form of guided iCBT that we have previously investigated[18] but represents a more modern iteration of the online platform with the same therapy form. The HUS-iCBT registry recorded patients from Dec 12, 2018, to Dec 22, 2022, and the FPQR from Aug 28, 2018, to Sept 28, 2022; however, all the patients in our research sample were registered at HUS before Jan 1, 2022. HUS was the only public-sector, nationwide iCBT provider, but it provided fCBT mainly for Uusimaa, the southwest region of Finland. For maximal comparability across different delivery formats for CBT for depression, we sampled patients from these registries according to the flowchart in figure 2A.

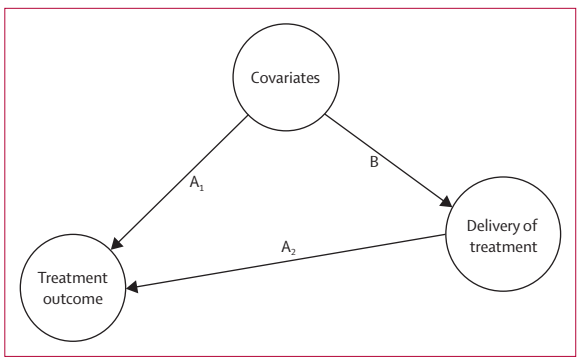

Figure 1: **Technical assumptions**
A_1, A_2, and B represent assumed dependency directions.

Each patient from both registries was linked with other national registries using their social-security identification number. The national Findata Data Permit Authority and the Digital and Population Services Agency linked the patients with their relatives (parents and siblings), after which both the patients and their relatives were linked with data from other registry owners (figure 2B). The registries contain accurate event dates for public-sector diagnoses and pharmaceutical purchases (eg, F codes and Anatomical Therapeutic Chemical-nervous system codes), but not private-sector diagnoses. Our research permission was granted by Findata according to national rules and regulations on secondary use of social and health data (permissions THL/4810/14.02.00/2020 and THL/1303/14.06.00/2023 to THR). These permissions relinquish the requirement for informed consent. This study was conducted in accordance with the Sixth World Medical Association Declaration of Helsinki and evaluated by the Ethics Committee of HUS (HUS/1861/2020 and HUS/3150/2020). The study was not pre-registered elsewhere.

Procedures

The people who had fCBT were listed in the FPQR and were undergoing primary-care and speciality-care CBT outsourced by public health care. All other treatment types within the FPQR were excluded from the study. We only included adults (ie, aged ≥18 years) listed on the

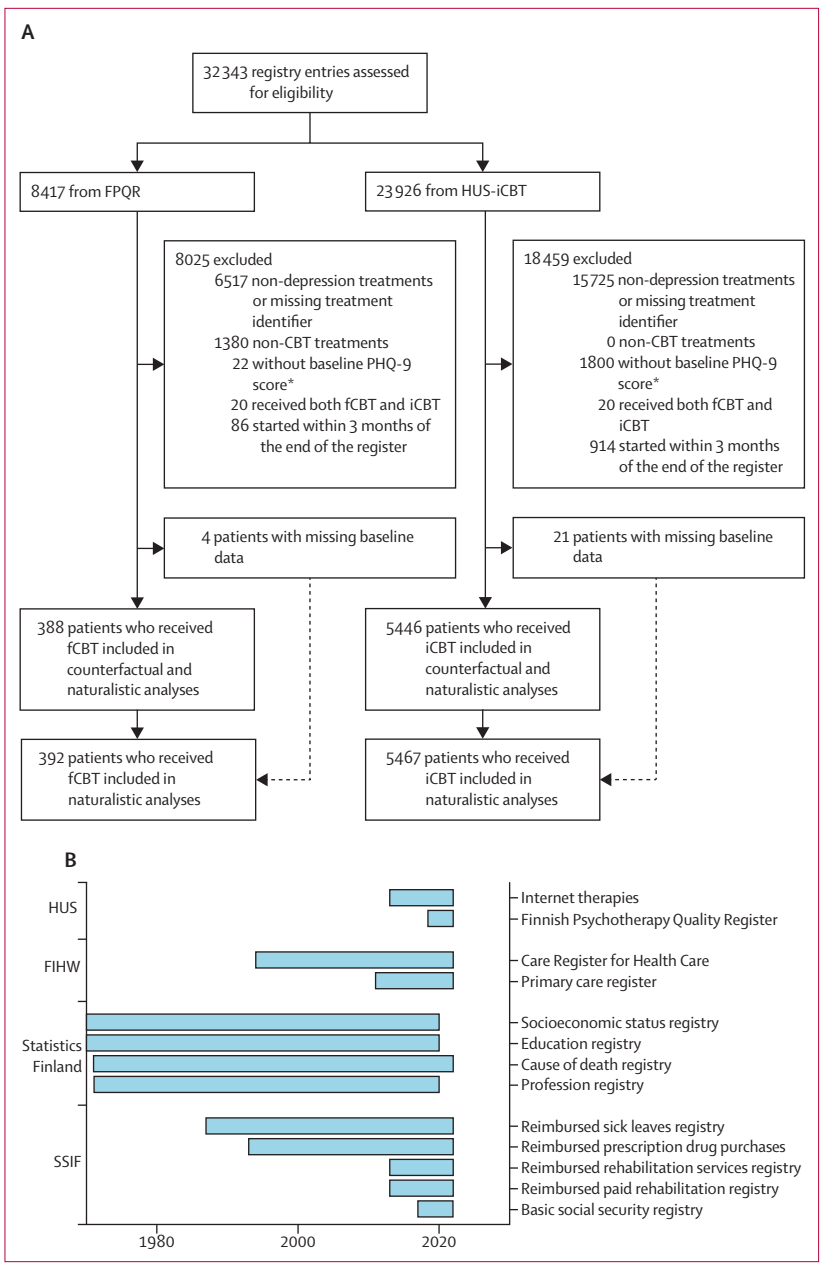

Figure 2: **Patient selection (A) and linked health and social security registries (B)**
(A) Flowchart of patient selection throughout the study. (B) Blue lines indicate time spans available for this study. Findata (and Digital and Population Services Agency) linked HUS-iCBT and fCBT registries with multiple registries from FIHW, Statistics Finland, and SSIF (also known as Kela). Outcome data and the covariate data of table 1 were extracted. CBT=cognitive behavioural therapy. fCBT=face-to-face CBT. FIHW=Finnish Institute of Health and Welfare. FPQR=Finnish Psychotherapy Quality Register. HUS-iCBT=HUS Helsinki University Hospital-owned iCBT. iCBT=internet-delivered CBT. PHQ-9=Patient Health Questionnaire-9. SSIF=Social Security Institute of Finland. *Duplicate entries to treatment of a same type also removed (fCBT or iCBT).

FPQR. Patient gender, referring to biological sex unless the individual has requested a legal change of gender, was retrieved from data available from the civil registry of Finland, with the options of male or female. We selected treatments of patients whose primary diagnosis for the treatment was depression (ICD-10 codes F32, F33, or F34.1) and when the therapist was trained in the CBT framework. All FPQR therapists have a 60-credit, 3-year training in their framework alongside their clinical work. The fCBT treatments typically contained up to 20 sessions altogether, but, in rare instances, could maximally contain 40 sessions per year, for up to 3 years.[16] A session was 45 min, and a conservative estimate of work time for each session would be 60 min, including preparation and documentation. With an observed mean number of 14 sessions in short FPQR treatments,[19] we estimated an 840-min average work-time expenditure per patient. Weekly sessions were recommended by the HUS Helsinki University Hospital.

The people given therapist-guided iCBT were listed in the HUS-iCBT registry. Patients entered iCBT with a doctor's prescription, with inclusion criteria for treatment being mild to moderate depression (F32.0 or F32.1), age over 15 years, and Finnish or Swedish language proficiency with motivation and technical means to work online independently. Exclusion criteria for receiving either fCBT or therapist-guided iCBT were acute suicide risk and current substance use. Further exclusion criteria for receiving iCBT were treatment-interfering psychotic or neurological behaviours, personality disorder, or chronic or bipolar depression. The suitability for either treatment was assessed by the attending physician. All guided iCBTs were delivered with the same online programme that aligns with CBT principles (including third-wave techniques such as mindfulness): the patient proceeds at their own pace while asynchronously talking with a trained therapist (eg, a psychologist or a nurse), who also provides unprompted guidance and an occasional telephone call (ideally one per patient per programme). The time spent by a therapist per patient as recommended by the HUS Helsinki University Hospital is 11 min per week, or 77 min in total for the seven weekly sessions. Patients were recommended by the HUS Helsinki University Hospital to complete one session per week and they could proceed more slowly or faster, up to one session per day. Not progressing in 2 weeks led to a call from the therapist and, ultimately, cessation of treatment if progress did not ensue despite encouragement.

In both treatment groups, we included treatments that had started 3 months before the end of our full sampling of the registries (Dec 31, 2021), thus giving time for the completion of therapy without completion being necessary. We only included patients for whom a baseline Patient Health Questionnaire (PHQ)-9 score was available. Both iCBT and fCBT were free to the patient, offered by the national public health-care system.

Choice of primary measure

As the primary outcome, we used the change in PHQ-9 score, a widely used and translated, brief, public-domain measure with good psychometric properties.[20,21] PHQ-9 scores range from 0 to 27, with higher scores indicating more severe or more frequent symptoms, and with a

decline of at least 1·7 points being subjectively detectable as the patient "feeling better".[22] We computed treatment response as the change in PHQ-9 score from baseline to the last measurement immediately after treatment, denoted change in PHQ-9; we did not focus on the automated follow-ups conducted 3 months after iCBT.

If a patient discontinued the treatment after providing only a single measurement, a change in PHQ-9 of 0 was assigned to that patient (proceeding in the iCBT programme required symptom reports; fCBT therapists were only advised to encourage patients to report). This conservative approach effectively penalises treatments with high dropout rates compared with complete-data or imputation analysis.

Statistical analysis

Both the fCBT and the iCBT registries also recorded overall anxiety severity and impairment scale scores, allowing us to use this continuous-valued score as a baseline predictor of the outcome.[23] Other predicting variables were drawn from the multiple linked administrative registries (figure 2B). For privacy and variance reduction, we only included categorical covariates that resulted in cell sizes ten observations or above when cross-tabulated with treatment allocation.

We arrived at our sample size by taking all eligible patients from the register. Our aim was to estimate the ATE between fCBT and guided iCBT. An estimate with a 95% CI no wider than 1·7 PHQ-9 points can be considered accurate in the sense of falling below a previously estimated detection threshold between "feeling better" versus "feeling the same" as before treatment.[22]

For TMLE, we implemented super learners with a broad set of base learners (examples 6 and 7 in Phillips and colleagues' article[24]), including generalised linear models, Bayesian generalised linear models, generalised additive models, lasso, elastic net and ridge regressions, polynomial splines, multivariate regression splines, extreme gradient boosting, Bayesian additive regression trees, and highly adaptive lasso.[24–27] TMLE requires two super learners, one predicting treatment allocation and another predicting treatment outcome.[13,14] Our super learners used cross-validation to automatically assign weight to the base learners with the non-negative least squares regression algorithm, with a guarantee of asymptotically better performance compared with the base learners.[13,24] We used the super learners to construct an estimate for ATE, using TMLE to optimise the bias-variance trade-off for the ATE parameter.[13,26] Thus, TMLE controlled for confounding by variables and their machine-learned, non-linear combinations and interactions. We followed the online and published guidance by Phillips and colleagues[24] and van der Laan and colleagues,[26] using their tlverse R environment, including packages sl3 (version 1.4.4), tmle3 (version 0.2.0), and cvAUC (version 1.1.4). This method implied nested ten-fold cross-validation for treatment outcome and

	Therapist-guided iCBT (n=5446)	fCBT (n=388)	p value (iCBT *vs* fCBT)
Continuous variables			
PHQ-9 score at first assessment*	13·31 (5·34)	11·57 (5·60)	<0·0001
OASIS score at first assessment	10·36 (3·68)	9·97 (3·81)	0·048
Distinct previous F-diagnoses, count	1·74 (2·61)	3·34 (3·92)	<0·0001
Distinct previous non-F-diagnoses, count	10·21 (9·55)	15·57 (12·55)	<0·0001
Relatives, parents plus siblings, count	4·08 (1·91)	3·78 (1·85)	0·0024
Maternal F-diagnoses, count	0·76 (2·29)	0·94 (2·83)	0·23
Paternal F-diagnoses, count	0·62 (2·10)	0·50 (1·94)	0·24
Siblings' F-diagnoses per sibling, count	0·58 (1·31)	0·50 (1·28)	0·27
Distinct purchased medicine (ATC-N) prescriptions, count	4·53 (4·20)	5·77 (5·52)	<0·0001
Distinct ATC-N medicine purchase events, count	26·77 (53·92)	45·00 (83·54)	<0·0001
Insurance-reimbursed sickness absence days, count	113·57 (187·58)	135·64 (196·74)	0·033
Insurance-reimbursed sickness absence days with F-diagnosis, count	71·17 (126·89)	72·80 (137·7)	0·82
Previous psychiatry visits, count	16·29 (41·45)	43·89 (81·63)	<0·0001
Age, years	34·49 (11·89)	40·37 (15·57)	<0·0001
Binary-valued variables			
Rehabilitation money received	25 (<1%)	12 (3%)	<0·0001
Income support received	1118 (21%)	71 (18%)	0·32
Rehabilitation support received	860 (16%)	59 (15%)	0·82
Rehabilitation psychotherapy received	502 (9%)	32 (8%)	0·58
Gender	..	..	0·070
Male	1634 (30%)	99 (26%)	..
Female	3812 (70%)	289 (74%)	..
First F-diagnosis for depression	1982 (36%)	140 (36%)	0·95
First F-diagnosis other†	259 (5%)	27 (7%)	0·069
Most frequent F-diagnosis for depression	2295 (42%)	185 (48%)	0·038
Most frequent F-diagnosis other†	1080 (20%)	78 (20%)	0·95
Siblings' most frequent F-diagnosis for depression	685 (13%)	39 (10%)	0·17
Siblings' most frequent F-diagnosis other‡	1447 (27%)	103 (27%)	1·00
Siblings' most frequent F-diagnosis for psychotic disorder	149 (3%)	12 (3%)	0·80
Mother's most frequent F-diagnosis for depression	469 (9%)	37 (10%)	0·60
Mother's most frequent F-diagnosis other†	772 (14%)	64 (16%)	0·24
Father's most frequent F-diagnosis for depression	281 (5%)	12 (3%)	0·093
Father's most frequent F-diagnosis other†	702 (13%)	50 (13%)	1·00
Most frequently purchased medicine was benzodiazepine	137 (3%)	18 (5%)	0·019
Most frequently purchased medicine was for migraine	240 (4%)	10 (3%)	0·11
Most frequently purchased medicine was non-SSRI for depression	945 (17%)	59 (15%)	0·31
Most frequently purchased medicine was paracetamol	799 (15%)	32 (8%)	0·0006
Most frequently purchased medicine was for psychosis	178 (3%)	10 (3%)	0·55
Most frequently purchased medicine was for sleep problems	157 (3%)	22 (6%)	0·0035

(Table 1 continues on next page)

	Therapist-guided iCBT (n=5446)	fCBT (n=388)	p value (iCBT *vs* fCBT)
(Continued from previous page)			
Most frequently purchased medicine was SSRI for depression	1821 (33%)	132 (34%)	0·89
Most frequently purchased medicine was opioid	461 (8%)	20 (5%)	0·028
Entrepreneur	186 (3%)	25 (6%)	0·0032
Upper white-collar employee	894 (16%)	58 (15%)	0·49
Lower white-collar employee	1774 (33%)	90 (23%)	<0·0002
Manual labourer	756 (14%)	39 (10%)	0·041
Student	953 (17%)	39 (10%)	<0·0002
Retiree	163 (3%)	68 (18%)	<0·0001
Unemployed or working status unknown	461 (8%)	46 (12%)	0·028
Native language is Finnish§	5092 (93%)	363 (94%)	1·00

Data are mean (SD) for continuous variables and frequency (%) for binary-valued variables. iCBT *vs* fCBT refers to a group comparison (p value from *t* test for continuous-valued variables and from χ^2 test for binary variables). Symptom scores are from the treatment registries, diagnoses refer to ICD-10 codes in the national Care Register maintained by Finnish Institute for Health and Welfare, work status is from Statistics Finland, demographic data from Digital and Population Data Services Agency, and the other variables from Social Security Institute of Finland (Kela). The variables were preprocessed to a format salient for the analyses. The variables (eg, F-codes) refer to the entire history available in the respective registries (figure 2B), with the exception that demographic, symptom, and work status refer to most recent values before treatment. The patients in the table correspond with those in the counterfactual analyses and exclude the 25 patients with some missing data (figure 2A; variable-specific missingness: 18 in gender and age, 21 in work status, four in OASIS, with some overlap). Whenever categories with <10 patients would have occurred, they were combined with other categories to retain patient anonymity and facilitate statistical inference. ATC-N=Anatomical Therapeutic Chemical-nervous system. CBT=cognitive behavioural therapy. iCBT=internet-delivered CBT. fCBT=face-to-face CBT. OASIS=Overall Anxiety Severity And Impairment Scale. PHQ-9=Patient Health Questionnaire-9. *Included only in the sensitivity analyses due to the artificial statistical dependence between the baseline and change scores in presence of measurement error. †Other F-diagnosis refers to those not in depression categories (F32, F33, F34.1; categories with <10 patients were combined). ‡Other F-diagnosis refers to those not in depression or psychotic categories (F32, F33, F34.1, F2, F39; categories with <10 patients were combined). §Most citizens, especially natively Finnish-speaking citizens, are of Finnic ethnic origin; routine collecting of data on race or ethnic origin is not currently permitted in Finland.

Table 1: **Descriptive statistics across the treatment groups**

nested stratified (by delivery format) ten-fold cross-validation for treatment allocation when investigating super-learner performance (nesting train-weight base learners cross-validation within select-validate super learner cross-validation; appendix 2 p 4). Non-nested (train-select) cross-validation was used within TMLE estimation (target parameter is a counterfactual quantity, precluding validation samples). We did multiple imputation using the superMICE package (version 1.1.1; five chains and five iterations). Learners, covariates, and imputation choices and examples of the central analysis scripts are further discussed in the appendix 2 (pp 2–12). We compared raw proportions using the χ^2 test and means using Welch's *t* test. Standardised mean difference effect sizes were given as Hedge's *g* and standardised change effects as Cohen's (paired) *d* effects.

See Online for appendix 2

As sensitivity analyses, we additionally computed the G-computation and AIPTW estimates of the ATE, and a version of TMLE with simple models.[14] We used linear regression as the outcome model and logistic regression for estimating the probability of receiving fCBT as opposed to iCBT. A previously recommended covariate balance investigation for (A)IPTW was used.[28] All the analyses took place in the Findata authority's secure Kapseli computing environment, in R (version 4.2.2).

Role of the funding source

The funder of the study had no role in study design, data collection, data analysis, data interpretation, or writing of the report.

Results

A total of 32 343 registry entries were assessed for eligibility, with 8417 coming from the FPQR and 23 926 from the HUS-iCBT registry. 8025 entries from the FPQR and 18 459 from the HUS-iCBT registry were excluded, resulting in a total of 392 patients from the FPQR, forming the fCBT group, and 5467 patients from the HUS-iCBT, forming the iCBT group, for naturalistic analyses (figure 2A). 25 patients were missing baseline data and were removed from counterfactual analyses for the primary outcome, resulting in 388 (99 [26%] male and 289 [74%] female) patients in the fCBT group and 5446 (1634 [30%] male and 3812 [70%] female) in the iCBT group (figure 2A). Four (<0·01%) of 40 263 patients whose data we accessed had objected to the use of their registry data by Findata (present inclusion criteria unknown; some of the accessed registers were not included in this study but could have contained the objecting four patients).

Four (1%) of 392 patients from the fCBT group and 21 (<1%) of 5467 patients from the iCBT group were missing mostly demographic baseline data. Table 1 summarises treatment group characteristics at baseline, after removing these 25 patients (other missingness pertaining to treatment dropout did not affect table 1). The mean age of patients in the final analytic fCBT group was 40 years (SD 16; median 36 years [IQR 28–51]) and 363 (94%) of 388 spoke Finnish as their native language. In the final guided iCBT group, the mean age was 34 years (SD 12; median 31 years [IQR 25–42]) with 5092 (93%) of 5446 speaking Finnish as their native language.

A mean of 9·4 weeks (SD 3·5) passed between baseline and the last follow-up assessment in guided iCBT for those who completed the last session and 7·7 weeks (4·4) for all patients (including patients whose first and last measurements were the same one). We estimated a mean interval of 35 weeks (22·9) between the reported start and end dates of the fCBT therapy period; this estimate was based on less accurate data reported to the outsourcing registry owner (in contrast, the iCBT programme automatically collects and dates data).

In the analytic iCBT group, there was a mean of 5·5 (SD 2·1; median 7·0 [IQR 4·0–7·0]) treatment sessions. In the fCBT group, reporting the number of sessions was not obligatory due to a coupon system that predefined the number of HUS-paid sessions; therefore, these data were only available for 352 (91%) of 388 patients (mean 12·3 [SD 9·8]; median 14·0 [IQR 2·0–20·0]). Of the 5467 people given iCBT, those who finished the treatment by reaching the seventh session (3333 [61%]) reported a mean decline

in PHQ-9 score of 4·8 points (SD 5·3; paired-sample Cohen's *d*=–0·91); symptoms increased (ie, PHQ-9 score increased) for 467 (14%) of 3333. The mean PHQ-9 decline defined via change in PHQ-9 in the full sample was 3·6 points (5·0; *d*=–0·71). However, the mean decline was 4·2 points (5·2; *d*=–0·81) when excluding 819 (15%) patients given iCBT who only reported the baseline PHQ-9 and therefore had null change in the score. If we were to include, as null changes, the 819 patients with only baseline PHQ-9 and the 1800 patients who were excluded due to being prescribed the programme but who never submitted a single PHQ-9 report (figure 2A), the PHQ-9 decline would have been 2·7 points (4·6; *d*=–0·58).

Of the 392 patients given fCBT, those who completed the PHQ-9 after treatment (316 [81%]) reported a mean PHQ-9 score decline of 3·0 points (SD 5·2; *d*=–0·59). For 72 (23%) of 316, symptoms increased; a greater proportion than the iCBT group (p<0·0001, χ^2=16·96). The full-data change in PHQ-9 was 2·4 points (4·8; *d*=–0·51). The raw iCBT–fCBT difference in this primary outcome metric in the total sample was statistically significant (1·1 points in favour of iCBT; *t*=–4·55, df 455·04, Hedge's *g*=–0·228 [95% CI –0·331 to –0·126]; p<0·0001). When including patients without the baseline PHQ-9 as not having changed score (413 patients for fCBT and 7194 for iCBT), the difference between the treatments was 0·4 points and not significant (*g*=–0·083 [–0·182 to 0·016]). In patients given iCBT, mean PHQ-9 scores declined less for male patients (3·1 points [SD 4·9]) compared with female patients (3·8 [5·0]; p<0·0001), but there was no gender difference among patients who completed iCBT treatment (4·5 points [5·4] in male patients *vs* 4·9 points [5·3] in female patients; p=0·095) or in patients who completed fCBT treatment (2·4 [4·5] *vs* 2·5 [4·9]; p=0·99).

The mean PHQ-9 score for both treatment groups was within the point range for moderate depression (table 1). Overall, patients with depression who were directed to fCBT differed from those directed to iCBT (third column of table 1). Thus, we considered what the treatment effect would be if all patients were given fCBT compared with giving them all iCBT by computing ATE estimates.

The super learners for treatment outcome and treatment assignment weighted partly distinct base learners (ie, algorithms) and designated as important only partly overlapping sets of variables (appendix 2 pp 13, 17). The cross-validated area-under-the-curve (AUC) performance measure for the model predicting treatment assignment to fCBT or iCBT was 0·77 (95% CI 0·74–0·79), meaning that the model was informative (a value of 0·5 indicates chance performance and 1·0 indicates perfect performance). The cross-validated coefficient of determination (R^2) for the outcome prediction model was 0·031 (~0·06 without cross-validation, or *r*=0·24). When we targeted the outcome and treatment-assignment models to ATE between fCBT versus iCBT, the result was close to the naturalistic treatment differential (table 2). In a scenario in which all the patients were assigned to the same treatment, if all patients were given iCBT the PHQ-9 change outcome would have been nearly the same, but slightly better, as it would have been if all were given fCBT; PHQ-9 score declined 0·745 points (95% CI 0·156–1·334) more in the therapist-guided iCBT group than in the fCBT group (table 2). The result was robust when restricting iCBT treatments to an fCBT delivery region (ie, restricting to residents of Uusimaa) and for alternative counterfactual definitions of dropout (table 2).

Recognising a particular black-box nature (appendix 2 pp 2–3) in the TMLE, we computed simpler estimates for ATE as sensitivity analyses. We took the ten most important predictors based on the super learners (appendix 2 p 17) and instead fitted a logistic regression model predicting treatment assignment and a linear regression model predicting change in PHQ-9 (appendix 2 pp 14–15). The assignment model performance was well above chance (cross-validated AUC 0·70 [95% CI 0·67 to 0·73]), although inferior to the super learner. The adjusted R^2 for the outcome prediction model was 0·03. After the removal of predictors that were non-significant in both the models (appendix 2 pp 14–15), their performance remained similar (cross-validated AUC 0·70 [0·67 to 0·73], adjusted R^2=0·03). Weighting by modelled inverse probabilities of treatment assignment resulted in an improved covariate balance across fCBT–iCBT samples (appendix 2 p 19). The finding of similar efficacies for iCBT and fCBT was robust

	Estimate (PHQ-9 points)*	SE of the estimate	95% CI of the estimate
All data (n=5834)			
Observed difference in means (change in PHQ-9 [fCBT]–change in PHQ-9 [iCBT])	1·120	0·312	0·509 to 1·731
Estimated ATE (change in PHQ-9 [fCBT])–(change in PHQ-9 [iCBT])	0·745	0·300	0·156 to 1·334
Uusimaa region only (n=3461)†			
Observed difference in means (change in PHQ-9 [fCBT]–change in PHQ-9 [iCBT])	1·045	0·334	0·391 to 1·700
Estimated ATE (change in PHQ-9 [fCBT]–change in PHQ-9 [iCBT])	0·636	0·326	–0·002 to 1·276
Estimated ATE if all had completed treatment (imputed; change in PHQ-9 [fCBT]–change in PHQ-9 [iCBT])	1·277	0·319	0·649 to 1·905
Estimated ATE if all had a second measurement (imputed; change in PHQ-9 [fCBT]–change in PHQ-9 [iCBT])	1·070	0·333	0·414 to 1·726

Patients with missing baseline data were not included in the analyses presented in this table. A negative change in PHQ-9 score indicates symptom reduction. ATE estimates were based on the targeted maximum likelihood estimator with super-learner machine learning. Results for all data and the Uusimaa region only set the change in PHQ-9 to 0 when the patient did not have a second measurement. Imputed results refer to counterfactuals based on multiple imputation of missing PHQ-9 reports: all data were used to impute all missing PHQ-9 data for sessions 3, 5, and 7 for all patients (including those given fCBT). In the first imputation-based analysis, the outcome was set to change from session 1 to 7. In the second analysis, the outcome was the same as in complete-data analyses (ie, from the first to last available measurement), but the imputed value from session 3 was inserted only for patients who did not have any follow-up measurements (in the complete-data analysis, their change was considered 0 instead). ATE=average treatment effect. CBT=cognitive behavioural therapy. fCBT=face-to-face CBT. iCBT=internet-delivered CBT. PHQ-9=Patient Health Questionnaire-9. *A positive estimate indicates greater symptom reduction in iCBT compared with fCBT. †Excludes patients from other regions, equivalent to 2373 patients given iCBT due to HUS Helsinki University Hospital Finnish Psychotherapy Quality Register only including fCBTs provided at Uusimaa.

Table 2: **Observed treatment effect comparison and estimated ATE (causal) differential**

to all the methodological choices: G-computed ATE 1·073, AIPTW-ATE 0·735 (95% CI 0·109 to 1·361), and simple TMLE-ATE 0·790 (0·164 to 1·415). Our prediction models did not use baseline PHQ-9 to avoid artificial dependence with outcome because it is part of the outcome already. A sensitivity analysis including baseline PHQ-9 in the assignment model had only a modest effect on estimates (G-computed ATE 1·073, AIPTW-ATE 0·128 [–0·593 to 0·850], and simple TMLE-ATE 0·114 [–0·569 to 0·797]). When including baseline PHQ-9 in both outcome and assignment models of full TMLE, the estimated ATE was –0·064 (–0·639 to 0·511).

Discussion

We found no mentionable difference between treatment effectiveness in guided iCBT versus fCBT for depression. This finding was consistent across raw data from hospital registries and multiple causal-effect estimates that retrospectively balanced the treatment groups. Our findings encourage wider dissemination of cost-effective and cost-efficacious guided iCBTs.

Our findings need to be interpreted with consideration of some limitations. First, the methods used achieve causal inference when relevant confounders are observed. Hence, we strived to include much registry-based background information and interpreted the results relative to RCTs. Nevertheless, variables missing from our set, such as digital affinity, could be of future interest, as could more accurate temporal and follow-up analyses. Second, the fCBT and guided iCBT treatments had qualitative differences other than delivery method. The fCBTs were an outsourced service from public mental health care, produced in private practices by licensed psychotherapists. In contrast, the trained iCBT therapists were employees in psychiatric specialty care (eg, psychologists and nurses), but not necessarily licensed psychotherapists. The fCBT therapists could adjust the treatment specifics during the treatment, whereas the online platform largely standardised the iCBT programme. On average, people having fCBT had more treatment sessions than those having iCBT. However, the number of sessions was found to be uncorrelated with effect sizes in past trials.[3] Third, the study was not pre-registered. However, sensitivity analyses showed a robustness of the results to many alternative analytical choices. Fourth, although HUS-iCBT clinical development extensively collaborates with individuals with lived experience, these individuals were not involved in this research specifically. Including their perspectives in future research might add valuable insights.

Finally, many more patients who were given guided iCBT dropped out from their treatment programme compared with the patients given fCBT. We accounted for these patients by using the last measurement taken before dropping out to compute the treatment outcome. Thus, iCBT was heavily penalised for the amount of dropout. Patients dropping out without a second measurement were recorded as having no change in their PHQ-9 score, which is lower than the average spontaneous recovery that occurs while on a waiting list.[29] Furthermore, one does not always attach a special clinical meaning to dropping out from treatment. For example, the stepped-care model might improve treatment outcomes by starting patients with a low-intensity treatment and by subsequently stepping those who do not respond to treatment up to a higher-intensity treatment.[30] We found that the average symptom change (both ATE and raw data) that included those who discontinued was no worse for iCBT than for fCBT, and iCBT had fewer treatment sessions in general. Thus, when dropout is not considered a primary indicator of treatment failure, iCBT provided similar treatment outcomes to fCBT, but reached these results more quickly.

Our study provides important additional support for guided iCBTs. Research has shown guided iCBTs to attain similar effects to fCBT in RCTs and in their meta-analyses (ie, efficacy).[1,2] We complemented this evidence by showing that an iCBT programme attained similar effects to fCBT also in naturalistic settings (ie, effectiveness) and when the naturalistic patients were statistically balanced across the treatments (ie, efficacy and effectiveness triangulated). The analytical frameworks of experiments (eg, RCTs) and counterfactual causal inference (eg, TMLEs) build on different assumptions but provide consistent evidence. Conceptually, such a situation of causal triangulation is considered the best-case scenario in modern causal-inference literature,[9–11] because guaranteeing the assumptions of any single method, including RCTs, is difficult, whereas guaranteeing that multiple methods did not all fail in distinct ways to provide the same, yet biased, finding is easier. If accepting equal treatment effects, a treatment that takes an average of 77 min of therapist time (iCBT) should be more cost-effective than a treatment that takes an average of 840 min of therapist time (fCBT); iCBT could therefore be 11-times more cost-effective.

Given the consistency of the present and past findings across various approaches to evaluating the efficacy of iCBT versus fCBT, our study offers reassurance regarding the validity of the analytical frameworks used. Similar formal methods for causal inference might represent cost-effective means for detecting active treatment components and predictors in mental health research in which confounding issues are pronounced.

We conclude that, for depressive disorders, evidence from RCTs, naturalistic studies, and quasi-experimental research all show similar treatment efficacy and effectiveness between guided iCBT and fCBT. Consequently, an internet delivery of short first-line treatments emerges as a cost-effective and scalable default treatment option. Nevertheless, maintaining a range of treatment alternatives remains important, considering the less-than-optimal efficacy of all available treatments,[8] special needs, and the emerging opportunities in treatment targeting.

Contributors

THR wrote the first draft of the report and conducted the data analyses. SES, SIS, JT, and J-HS wrote critical revisions to the manuscript and accepted the final version. THR, SES, and JT have accessed and independently verified the research data, the collection of which was monitored by the administrative personnel at Findata and HUS Tietohallinto; respective registry owners have verified the original registries. Only Findata authority had access to the full personal data, with THR, SES, and JT only accessing pseudonymised research data, preprocessed by the Findata authority after consultation of the national Office of the Data Protection Ombudsman. THR and JT independently verified and re-computed results for all the naturalistic observations. THR conducted the machine learning, with the code being in the appendix 2 (pp 22–29). THR and JT did the data curation and formal analysis. THR, SES, and J-HS did the investigation. THR created the methodology and did the project administration. THR, SES, and J-HS found resources. THR and JT identified software and wrote scripts. THR and SES supervised the project. THR did the project visualisation. All authors reviewed and edited the manuscript and performed the literature search. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

Declaration of interests

We declare no competing interests.

Data sharing

As per Finnish legislations, pseudonymised research data are available to researchers granted permission from the Findata Social and Health Data Permit Authority (https://findata.fi/en/), whose activities are based on the Act on the Secondary Use of Health and Social Data. The authors do not have permission to share data with third parties directly. The analysis scripts are available in the appendix 2 (pp 22–29).

Acknowledgments

This study was supported by the Research Council of Finland (also known as the Academy of Finland) with grant numbers 334057, 335901, and 358138 given to THR and the Machine-learned causal factors for psychotherapy project. We thank Ville Ritola for comments on an earlier version of the manuscript. SES was supported by Social Insurance Institution of Finland (Kela 140/331/2021).

References

1 Hedman-Lagerlöf E, Carlbring P, Svärdman F, Riper H, Cuijpers P, Andersson G. Therapist-supported internet-based cognitive behaviour therapy yields similar effects as face-to-face therapy for psychiatric and somatic disorders: an updated systematic review and meta-analysis. *World Psychiatry* 2023; **22:** 305–14.

2 Andersson G, Cuijpers P, Carlbring P, Riper H, Hedman E. Guided internet-based *vs* face-to-face cognitive behavior therapy for psychiatric and somatic disorders: a systematic review and meta-analysis. *World Psychiatry* 2014; **13:** 288–95.

3 Ciharova M, Karyotaki E, Miguel C, et al. Amount and frequency of psychotherapy as predictors of treatment outcome for adult depression: a meta-regression analysis. *J Affect Disord* 2024; **359:** 92–99.

4 Pontén M, Jonsjö M, Vadenmark V, et al. Association between expectations and clinical outcomes in online *v* face-to-face therapy—an individual participant data meta-analysis. *Psychol Med* 2024; **54:** 1207–14.

5 Moshe I, Terhorst Y, Philippi P, et al. Digital interventions for the treatment of depression: a meta-analytic review. *Psychol Bull* 2021; **147:** 749–86.

6 Tröger A, Miguel C, Ciharova M, et al. Baseline depression severity as moderator on depression outcomes in psychotherapy and pharmacotherapy. *J Affect Disord* 2024; **344**: 86–99.

7 van der Lem R, de Wever WW, van der Wee NJ, van Veen T, Cuijpers P, Zitman FG. The generalizability of psychotherapy efficacy trials in major depressive disorder: an analysis of the influence of patient selection in efficacy trials on symptom outcome in daily practice. *BMC Psychiatry* 2012; **12:** 192.

8 Ormel J, Hollon SD, Kessler RC, Cuijpers P, Monroe SM. More treatment but no less depression: the treatment-prevalence paradox. *Clin Psychol Rev* 2022; **91:** 102111.

9 Hammerton G, Munafò MR. Causal inference with observational data: the need for triangulation of evidence. *Psychol Med* 2021; **51:** 563–78.

10 Lawlor DA, Tilling K, Davey Smith G. Triangulation in aetiological epidemiology. *Int J Epidemiol* 2016; **45:** 1866–86.

11 Dahabreh IJ, Bibbins-Domingo K. Causal inference about the effects of interventions from observational studies in medical journals. *JAMA* 2024; **331:** 1845–53.

12 Hernán MA, Robins JM. Causal inference: what if. CRC Press, 2024.

13 van der Laan MJ, Rose S. Targeted learning: causal inference for observational and experimental data. Springer, 2011.

14 Luque-Fernandez MA, Schomaker M, Rachet B, Schnitzer ME. Targeted maximum likelihood estimation for a binary treatment: a tutorial. *Stat Med* 2018; **37:** 2530–46.

15 Torvik FA, Ystrom E, Gustavson K, et al. Diagnostic and genetic overlap of three common mental disorders in structured interviews and health registries. *Acta Psychiatr Scand* 2018; **137:** 54–64.

16 Saarni SE, Rosenström TH, Stenberg J-H, et al. Finnish Psychotherapy Quality Register: rationale, development, and baseline results. *Nord J Psychiatry* 2023; **77:** 455–66.

17 Ritola V, Lipsanen JO, Pihlaja S, et al. Internet-delivered cognitive behavioral therapy for generalized anxiety disorder in nationwide routine care: effectiveness study. *J Med Internet Res* 2022; **24:** e29384.

18 Rosenström TH, Ritola V, Saarni S, Joffe G, Stenberg J-H. Measurement invariant but non-normal treatment responses in guided internet psychotherapies for depressive and generalized anxiety disorders. *Assessment* 2023; **30:** 618–32.

19 Rosenström TH, Mylläri S, Malkki V, Saarni SE. Feasibility of generic, short, and easy-to-use assessment of psychological distress during psychotherapy: longitudinal measurement invariance of CORE-10 and -OM. *Psychother Res* 2022; **32:** 1090–99.

20 Kroenke K, Spitzer RL, Williams JB. The PHQ-9: validity of a brief depression severity measure. *J Gen Intern Med* 2001; **16:** 606–13.

21 Stochl J, Fried EI, Fritz J, et al. On dimensionality, measurement invariance, and suitability of sum scores for the PHQ-9 and the GAD-7. *Assessment* 2022; **29:** 355–66.

22 Kounali D, Button KS, Lewis G, et al. How much change is enough? Evidence from a longitudinal study on depression in UK primary care. *Psychol Med* 2022; **52:** 1875–82.

23 Norman SB, Cissell SH, Means-Christensen AJ, Stein MB. Development and validation of an Overall Anxiety Severity And Impairment Scale (OASIS). *Depress Anxiety* 2006; **23:** 245–49.

24 Phillips RV, van der Laan MJ, Lee H, Gruber S. Practical considerations for specifying a super learner. *Int J Epidemiol* 2023; **52**: 1276–85.

25 Hastie T, Tibshirani R, Friedman J. The elements of statistical learning: data mining, inference, and prediction, 2nd edn. Springer-Verlag, 2009.

26 van der Laan M, Coyle J, Hejazi N, Malenica I, Phillips R, Hubbard A. Targeted learning in R: causal data science with the tlverse software ecosystem. Aug 10, 2022. https://tlverse.org/tlverse-handbook/ (accessed April 17, 2024).

27 Benkeser D, van der Laan M. The highly adaptive Lasso estimator. IEEE International Conference on Data Science and Advanced Analytics; Oct 17–19, 2016: 689–96.

28 Chesnaye NC, Stel VS, Tripepi G, et al. An introduction to inverse probability of treatment weighting in observational research. *Clin Kidney J* 2022; **15:** 14–20.

29 Whiteford HA, Harris MG, McKeon G, et al. Estimating remission from untreated major depression: a systematic review and meta-analysis. *Psychol Med* 2013; **43:** 1569–85.

30 van Straten A, Hill J, Richards DA, Cuijpers P. Stepped care treatment delivery for depression: a systematic review and meta-analysis. *Psychol Med* 2015; **45:** 231–46.